RAJALAKSHMI ENGINEERING COLLEGE
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AD23532 – PRINCIPLES OF DATA SCIENCE
EXPERIMENT 3A: MODEL PLANNING AND BUILDING – CLASSIFICATION

QUESTION:
A telecommunications company wants to predict customer churn using machine learning classification models. Train and compare Logistic Regression, Decision Tree, and Random Forest classifiers using the preprocessed dataset. Evaluate their performance using classification metrics, identify the best-performing model, and recommend strategies to reduce customer churn.

AIM:
To build and evaluate Logistic Regression, Decision Tree, and Random Forest classification models using the CustomerChurn_Preprocessed.csv dataset, compare their performance using a confusion matrix, accuracy, precision, recall, and F1-score, identify the best-performing model, and provide business recommendations to reduce customer churn.

ALGORITHM:
1. Import the required Python libraries for data handling, visualization, model building, and evaluation.
2. Load the CustomerChurn_Preprocessed.csv dataset and separate the input features and target variable Churn.
3. Encode categorical features and split the dataset into 80% training data and 20% testing data.
4. Train a Logistic Regression classifier and predict customer churn on the test dataset.
5. Train a Decision Tree classifier and predict customer churn on the test dataset.
6. Train a Random Forest classifier and predict customer churn on the test dataset.
7. Evaluate all three models using confusion matrices, accuracy, precision, recall, and F1-score.
8. Compare the performance of the three models using a bar chart and identify the best-performing model.
9. Provide three business recommendations based on the model results to help reduce customer churn.

PROGRAM & OUTPUT:

```
# Name: Shyam M (241801267)
# Section 1: Import Required Libraries

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    confusion_matrix, accuracy_score, precision_score,
    recall_score, f1_score, classification_report
)
```

```
# Name: Shyam M (241801267)
# Section 2: Load Dataset and Prepare Features

df = pd.read_csv("/content/WA_Fn-UseC_-Telco-Customer-Churn.csv")

X = df.drop(columns=["Churn", "customerID"], errors="ignore")
y = df["Churn"].map({"No": 0, "Yes": 1})

X = pd.get_dummies(X, drop_first=True, dtype=int)
X = X.fillna(X.median(numeric_only=True)).fillna(0)
```

```
# Name: Shyam M (241801267)
# Section 3: Split Training and Testing Data

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
```

```
# Name: Shyam M (241801267)
# Section 4: Build Classification Models

models = {
    "Logistic Regression": make_pipeline(
        StandardScaler(),
        LogisticRegression(max_iter=1000)
    ),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(
        n_estimators=100, random_state=42
    )
}

results = {}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    results[name] = {
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred, zero_division=0),
        "Recall": recall_score(y_test, y_pred, zero_division=0),
        "F1-Score": f1_score(y_test, y_pred, zero_division=0)
    }
```

```
# Name: Shyam M (241801267)
# Section 5: Evaluate Models

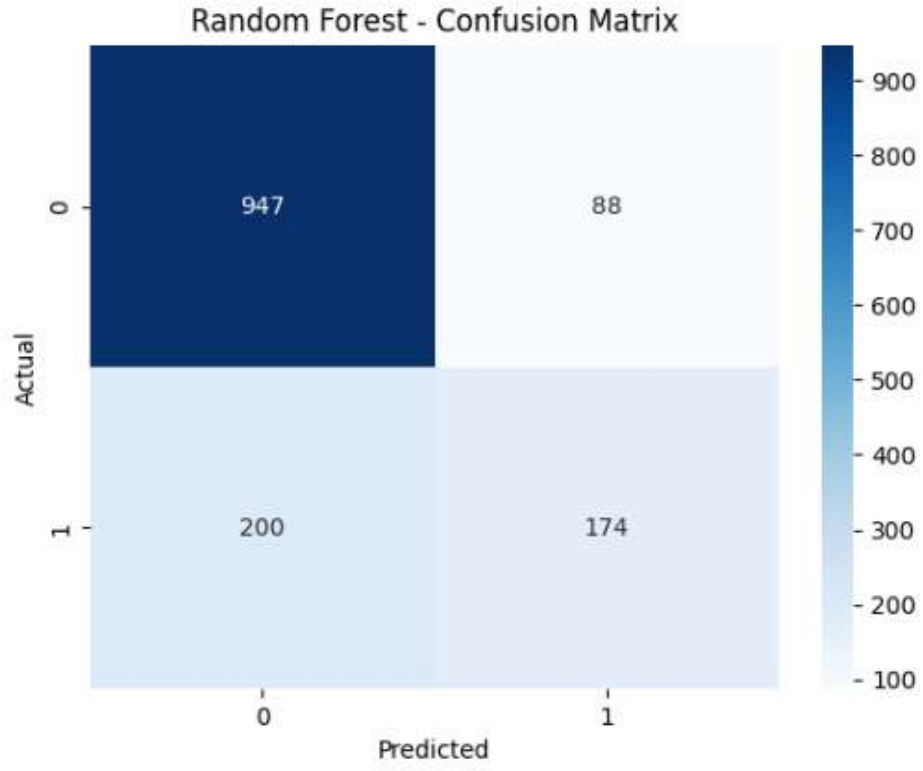
print("\nModel:", name)
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred, zero_division=0))

sns.heatmap(
    confusion_matrix(y_test, y_pred),
    annot=True, fmt="d", cmap="Blues"
)
plt.title(name + " - Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

```
Model: Random Forest
Confusion Matrix:
 [[947  88]
 [200 174]]
              precision    recall  f1-score   support

           0       0.83      0.91      0.87      1035
           1       0.66      0.47      0.55       374

    accuracy                           0.80      1409
   macro avg       0.74      0.69      0.71      1409
weighted avg       0.78      0.80      0.78      1409
```

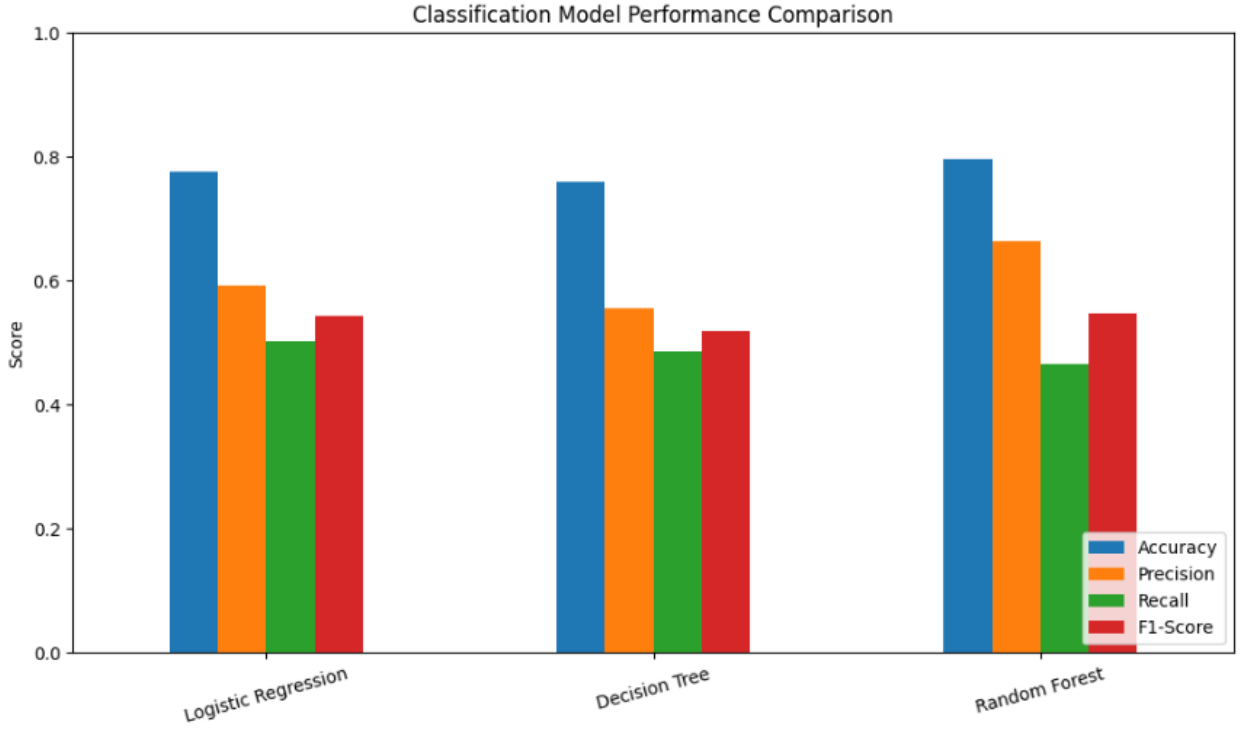


```
# Name: Shyam M (241801267)
# Section 6: Compare Model Performance

scores = pd.DataFrame(results).T
print("\nModel Comparison:\n", scores)

scores.plot(kind="bar", figsize=(10, 6))
plt.title("Classification Model Performance Comparison")
plt.ylabel("Score")
plt.ylim(0, 1)
plt.xticks(rotation=15)
plt.legend(loc="lower right")
plt.tight_layout()
plt.show()
```

```
Model Comparison:
                     Accuracy  Precision    Recall  F1-Score
Logistic Regression  0.776437   0.593060  0.502674  0.544139
Decision Tree        0.760114   0.554878  0.486631  0.518519
Random Forest        0.795600   0.664122  0.465241  0.547170
```



```
# Name: Shyam M (241801267)
# Section 7: Identify Best Model

best_model = scores["F1-Score"].idxmax()
print("Best Model based on F1-Score:", best_model)
```

```
Best Model based on F1-Score: Random Forest
```

```
# Name: Shyam M (241801267)
# Section 8: Business Recommendations

print("\nBusiness Recommendations:")
print("1. Offer retention plans to customers predicted to churn.")
print("2. Improve customer support and service quality.")
print("3. Provide suitable discounts and flexible contract options.")
```

```
Business Recommendations:
1. Offer retention plans to customers predicted to churn.
2. Improve customer support and service quality.
3. Provide suitable discounts and flexible contract options.
```

RESULT:

Thus, Logistic Regression, Decision Tree, and Random Forest classification models were trained and evaluated using the Telco Customer Churn dataset. Their performance was compared using a confusion matrix, accuracy, precision, recall, and F1-score. The best-performing model was identified, and business recommendations were provided to help the telecommunications company reduce customer churn.